



WaterWorks TON test strip accuracy

A report prepared on behalf of Wye Salmon Association (WSA)

A quick evaluation of WaterWorks nitrate test strips using a Hach standard solution.

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Introduction

Wye Alliance citizen scientists have historically used the Hach AquaChek TON/nitrite test strips for water quality checks. These have proved robust and accurate but have recently been removed from the UK market due to chemical regulatory issues. Initially, an alternative product from Simplex Health was proposed but early use suggested that these might be inferior. A bench comparison performed in November-25, using test solutions derived from a Hach nitrate calibration solution, confirmed the poorer performance. Simplex Health test strips were found to have relatively weak colour development – essentially reading consistently low throughout their range (0 – 50mgN/l).

Much better performance has been obtained with a product from WaterWorks (0-50mgN/l), which has now been adopted. The aim of this work was to complete the picture by performing a similar bench evaluation using the Hach nitrate calibration solution.

Method

As in the previous trial, a standard solution of nitrate having a concentration of 100 mgN/l (Hach cat. 194749) was used to prepare 6 test solutions in separate beakers by stepwise dilution with DI water (1-5 $\mu\text{S}/\text{cm}$) using a measuring cylinder. Concentrations were chosen to correspond to colours on the WaterWorks colour chart, as per table 1.

Concentration (mgN/l)	Preparation
50	50% dilution of stock solution
20	20% dilution of stock solution
10	50% dilution of 20 mgN/l
5	50% dilution of 10 mgN/l
2	40% dilution of 5 mgN/l
0.5	25% dilution of 2 mgN/l

Table 1: test solution preparation

At the time of testing, the solution temperatures were all 19°C.

Following instructions on the packaging, WaterWorks test strips were immersed for 2 seconds and developed for 60 seconds. Colour development was rapid, similar to the Hach test strips evaluated previously (however, colour development could be slower at low temperatures).

Colour comparison at 60 seconds was found to be accurate and unambiguous at each concentration.

As for all test strips, colours fade as the pad dries out, so prompt reading is essential.

Expiry date for the WaterWorks test strips was April-27.

Conclusion

- WaterWorks test strips provide an accurate colour match for TON throughout the range of 0.5, 2, 5, 10, 20 & 50 mgN/l of nitrate.
- They are therefore an acceptable replacement for Hach AquaChek strips.